

# XQdrant: In-Database Attribution and Masked-Distance Subspace Search over HNSW

Anonymous Author(s)

Anonymous Institution    `anonymous@domain.edu`

Draft — July 2026

## Abstract

Modern vector databases return opaque scalar similarities over high-dimensional embeddings, leaving practitioners without coordinate-level explanations and without a way to *route* HNSW search using only a dynamic focus subspace. We present **XQdrant**, an extension of the Qdrant engine with two complementary capabilities. First, `with_dims_explained` elevates exact metric score decomposition to a first-class, in-database post-search primitive, avoiding client-side vector egress; on synthetic encoder-width corpora this yields about  $2.2\times$  lower attribution latency than post-query extraction at matched Recall@10. Second, we study *masked-distance* HNSW traversal via `focus.masked`, a hybrid layer cutoff (`mask_from_layer`), SIMD gather/repack kernels, an optional verify/overfetch pass, and a weighted score blend (`alpha`). Across SIFT1M and high-dimensional synthetic indices we find that hybrid cutoffs recover navigability (SIFT recall  $\sim 0.8\text{--}0.97$  for cutoffs  $\geq 1$ ) that naive all-layer masking collapses, that gather kernels win microbenchmarks ( $\sim 1.1\text{--}1.8\times$ ) but do not unlock end-to-end speedups vs. full search, that subspace $\leftrightarrow$ full  $k$ -NN coherence predicts when verification can help, and that plain verify at *limit=k* cannot change Recall@ $k$  whereas overfetch+verify recovers full-space recall on coherent data (SIFT) at substantial latency cost. A weighted blend is dominated by a hard layer cutoff. We release an experiment index and reproducible harness so these negative and positive results can be cited independently.

**Keywords:** vector databases, HNSW, approximate nearest neighbor, explainability, feature attribution, subspace search, masked distance, SIMD

## 1 Introduction

High-dimensional dense embeddings underpin retrieval-augmented generation (RAG), recommendation, and semantic search [6, 4, 3]. Production engines typically traverse HNSW graphs [1], collapsing each comparison into a single scalar. That design creates two gaps. *Explainability*: practitioners cannot see which coordinates drove a hit without fetching full vectors and re-computing per-dimension terms outside the database. *Subspace routing*: many workloads care about a dynamic focus set  $\mathcal{F}$  of coordinates (concept probes, ablations, modality masks), yet stock HNSW always evaluates full  $D$ -dimensional distances during graph descent. Preselect-and-rescore on  $\mathcal{F}$  improves semantic specificity but cannot reduce traversal work [12].

**XQdrant** addresses both gaps inside a Qdrant-compatible Rust engine. For attribution, `with_dims_explained` attaches exact top- $m$  coordinate terms after search. For subspace routing, `nearest.focus` with `masked=true` evaluates reduced distances during HNSW, optionally with a hybrid layer cutoff (`mask_from_layer`), gather-based SIMD kernels, a verify/overfetch rescoring mode, and an  $\alpha$ -blend of full and masked scores.

**Contributions.**

**Algorithm 1** (In-database top- $m$  attribution).

**Input:** query  $\mathbf{q}$ , vector  $\mathbf{v}$ ,  $D$ ,  $m$ , metric  $\delta$ .

1. Normalize operands if required by  $\delta$ .
2. For  $i = 0, \dots, D - 1$ : append  $(i, \text{TERM}_i(\mathbf{q}, \mathbf{v}, \delta))$ .
3. Return order-statistic top- $m$  by  $|t|$ , sorted descending.

Figure 1: In-database top- $m$  feature attribution.

- **In-database attribution.** Opt-in exact score decomposition with order-statistic top- $m$  selection; recall-neutral vs. vanilla Qdrant and  $\sim 2.2\times$  faster than post-query extraction at encoder widths.
- **Masked-distance HNSW.** Layer-aware dual scorers, API-stable focus fields (`masked`, `mask_from_layer`, `verify`, `alpha`), and gather/repack kernels selectable at runtime.
- **Empirical map of what works.** Hybrid cutoffs restore navigability on SIFT; kernels win microbenches but not wall-clock search; coherence explains high- $D$  plateaus; verify needs overfetch;  $\alpha$ -blend is a negative ablation vs. a hard cutoff.
- **Reproducible artifacts.** Scripts under `DB.systems/research/` and a curated experiment index listing canonical CSVs/plots for each claim.

## 2 Background

### 2.1 HNSW and Contiguous SIMD

HNSW maintains multi-layer proximity graphs [1]. Queries greedily descend with candidate heap size `ef_search`, then extract top- $k$  at the base layer. Distance kernels assume contiguous `f32` slices (AVX2/NEON), so a non-contiguous focus set either (i) repacks selected coordinates into a scratch buffer or (ii) gathers them in-register/*via* gather loads.

### 2.2 Score Decomposition

For Dot product,  $s(\mathbf{q}, \mathbf{v}) = \sum_i q_i v_i$ ; Cosine applies the same after  $\ell_2$  normalization; Euclidean/Manhattan admit analogous additive terms [7, 8]. Attribution is thus exact accounting of the engine’s own metric—not a surrogate XAI model.

### 2.3 Subspace Coherence

Let  $N_k^{\text{full}}(q)$  and  $N_k^{\mathcal{F}}(q)$  be exact  $k$ -NN sets under the full metric and under  $\mathcal{F}$ . *Coherence*  $|N_k^{\text{full}} \cap N_k^{\mathcal{F}}|/k$  predicts when masked routing can surface full-space neighbors and when a cheap rescore can help.

## 3 In-Database Attribution

XQdrant injects attribution at the collection coordinator after shard merge, with an early return when `with_dims_explained` is absent. Per explained hit, Algorithm 1 materializes  $D$  terms and selects the top- $m$  by absolute contribution in average  $\mathcal{O}(D)$  time (`select_nth_unstable`).

`nearest.focus` (without masking) still performs full HNSW preselection then rescoring on  $\mathcal{F}$ —useful for interpretability-constrained ranking but not a traversal acceleration (Section 5.2).

## 4 Masked-Distance HNSW Design

### 4.1 API Surface

Focus parameters ride inside the existing query body:

```
"focus": {  
  "dims": [...], "masked": true,  
  "mask_from_layer": 1,    // optional hybrid cutoff  
  "verify": true,          // optional full rescore  
  "alpha": 0.25            // optional blend on focus layers  
}
```

Constraints mirror dense local search: no quantization/turbo, no simultaneous `with_dims_explained` on masked paths, fail-fast if `mask_from_layer/verify/alpha` are set without `masked=true`.

### 4.2 Option 1: All-Layer Masking

With `masked=true` and cutoff omitted, every HNSW comparison uses only coordinates in  $\mathcal{F}$ . This is the cheapest semantic change but destroys navigability when the focus metric disagrees with the index geometry.

### 4.3 Option 2: Hybrid Layer Cutoff

Layer  $\ell$  uses the masked scorer iff  $\ell < L$  for cutoff  $L=\text{mask\_from\_layer}$ ; otherwise full distance.  $L=0$  routes to plain nearest (no masking); a large  $L$  recovers Option 1. Implementation builds dual raw scorers inside `FilteredScorer` and switches with `set_layer` at layer boundaries—visited-set and heap logic remain unchanged.

### 4.4 Kernels: Repack vs. Gather (X1)

Masked scoring historically copied focus dims into a scratch buffer then called contiguous SIMD (*repack*). We add an in-place *gather* path (AVX2 `vgatherdps` for Dot/Cosine/Euclid on x86; scalar elsewhere), selected by `XQDRANT_MASKED_KERNEL`.

### 4.5 Verify and Overfetch (X3)

`verify=true` re-scores returned candidates with pure full distance after traversal (same hook as quantization rescore). At *limit=k* this only reorders a size- $k$  set, so  $\text{Recall}@k$  is unchanged. Overfetch (*limit=m·k*, keep top- $k$ ) turns masked search into a candidate filter.

### 4.6 Weighted Blend (Option 4)

On focus layers,  $\text{score} = \alpha \cdot s_{\text{full}} + (1 - \alpha) \cdot s_{\text{focus}}$ . Coarse hybrid layers stay full-only. Blend alone never reduces full-distance work; it only biases routing.

## 5 Experimental Evaluation

### 5.1 Setup

**Attribution suite.** Synthetic corpora  $N=10k$ ,  $Q=500$ ,  $D \in \{768, 1536\}$ , seed 42; baselines: vanilla Qdrant, post-query extract+client top- $m$ , and XQdrant. Hardware for those runs: Apple Silicon laptop (suite orchestrator).

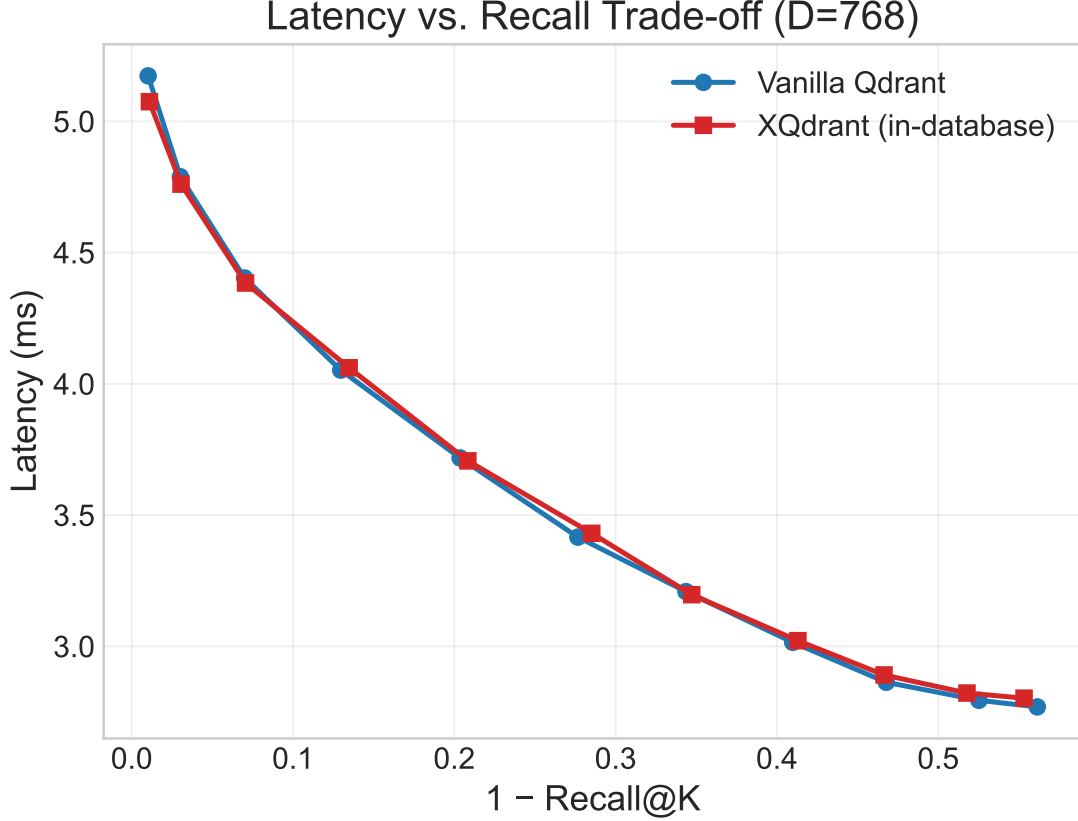


Figure 2: Attribution suite: latency–recall at  $D=768$  with `with_dims_explained` ( $m=10$ ).

**Masked-distance suite.** HTTP against a release XQdrant binary, typically with `XQDRANT_MASKED_KERNEL=g`. Corpora: SIFT1M ( $D=128$ ) and synthetic  $N=50k$  at  $D \in \{768, 1536\}$ ,  $Q=500$ ,  $k=10$ , `ef_search=128`. Canonical folders are listed in `DB.systems/experiments/README.md`; figures below are copied into `paper/figures/`.

## 5.2 Attribution and Focus Rescoring

Figure 2 shows that enabling `with_dims_explained` preserves the latency–recall frontier vs. vanilla Qdrant at  $D=768$  (Recall@10 gap  $< 0.01$ ). Figure 3: in-database attribution holds  $\sim 3.5$  ms p50 across  $m \in \{1, \dots, 50\}$  while post-query extraction sits near 7.8 ms ( $\approx 2.2\times$ ). Figure 4: preselect-and-rescore focus yields speedup  $< 1$  (0.84–0.87), motivating masked traversal rather than rescoring alone.

## 5.3 Option 1: Naive Masking Collapses Full Recall

Figure 5: on SIFT1M, all-layer masked search vs. *full* ground truth yields Recall@10  $\approx 0.006/0.079/0.257/0.464/$  at ratios 0.1/0.25/0.5/0.75/1.0.

## 5.4 Option 2: Hybrid Cutoff Recovers Navigability

Figure 6: with gather kernels, SIFT hybrid cutoffs  $L \geq 1$  recover  $\sim 0.79$ – $0.98$  recall (script subspace/hybrid GT), while  $L=0$  (plain nearest vs. that GT) tracks low coherence. High- $D$  synthetic hybrids lift collapsed cells (e.g.,  $D=768$ , ratio 0.25:  $\sim 0.01 \rightarrow \sim 0.20$ ) but plateau near 0.15–0.22. **End-to-end speedup vs. full search remains  $< 1$**  in every gathered cell (Figure 7).

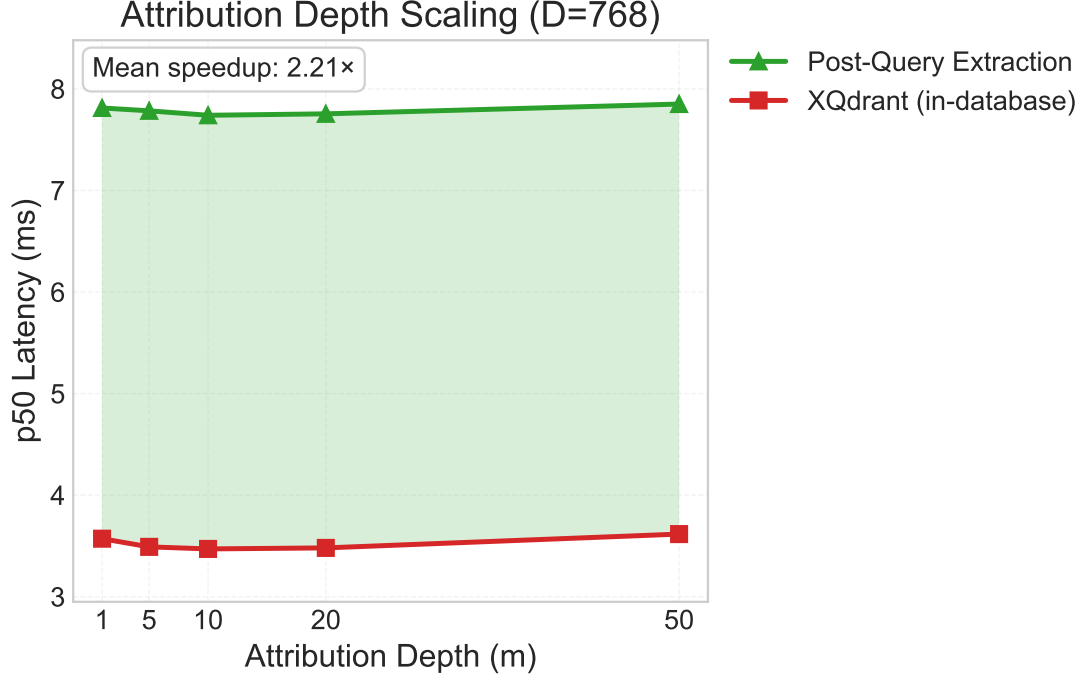


Figure 3: Attribution depth: in-DB vs. post-query extraction ( $D=768$ ).

### 5.5 X1: Gather Wins Microbenchmarks, Not Wall-Clock

Figure 8: gather is  $\sim 1.24\times/1.14\times/1.42\times/1.77\times$  faster than repack at  $D_{\text{sub}} \in \{32, 128, 384, 768\}$  (ns/op). Re-running Option 2 with gather leaves recall intact but still shows **zero** end-to-end cells with speedup  $> 1$ .

### 5.6 X2: Coherence Predicts High- $D$ Failure

Figure 9: overlap@10 grows with ratio but stays low on synthetic  $D=768$  ( $\sim 0.007 \rightarrow 0.30$  from ratio  $0.1 \rightarrow 0.75$ ); SIFT is somewhat higher ( $\sim 0.01 \rightarrow 0.46$ ). This explains why hybrid navigability does not imply full-space recall recovery.

### 5.7 X3: Verify Needs Overfetch; Costly When It Helps

Evaluating against **full-space** GT (corrected harness): at overfetch  $m=1$ , masked and verify Recall@10 are identical (same  $k$  IDs). At  $m=8$  on SIFT with hybrid  $L=1$ , full recall rises e.g.  $0.26 \rightarrow 0.66$  (ratio 0.5) and  $0.45 \rightarrow 0.89$  (ratio 0.75), but p50 jumps by  $\sim +45$  ms ( $\sim 1.5 \rightarrow \sim 46$  ms). High- $D$  synthetic stays  $\leq \sim 0.19$  even with  $m=8$  (Figure 10).

### 5.8 Option 4: Blend Is Dominated by $\alpha=0$

Figure 11: at ratio 0.25, SIFT hybrid recall is best at  $\alpha=0$  ( $\sim 0.84$  for  $L \geq 1$ ) and collapses monotonically toward the coherence floor as  $\alpha \rightarrow 1$ . Hard `mask_from_layer` without blend is strictly preferable.

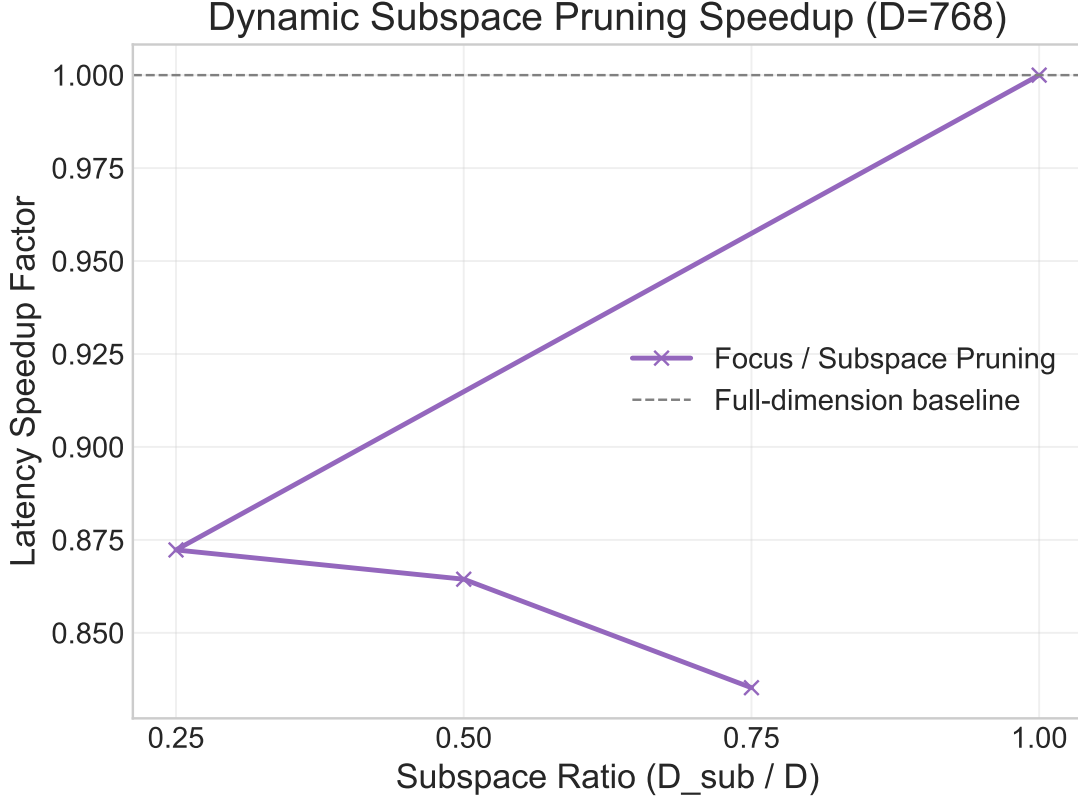


Figure 4: Focus *rescore* speedup  $T_{\text{full}}/T_{\text{sub}}$  ( $D=768$ ): values  $< 1$  show rescoring does not accelerate search.

## 5.9 Summary Table

## 6 Related Work

**Vector databases.** Milvus [3], Pinecone [5], and Qdrant [4] expose ANN with filters and quantization but not native per-coordinate decompositions or layer-aware masked metrics.

**ANN libraries.** FAISS [2] and ScaNN [9] optimize large-scale search; explainability and dynamic subspace routing remain post-processing.

**Explainable AI.** SHAP/LIME [7, 8] approximate model behavior. We instead exploit additive metric structure for deterministic attributions aligned with ranking, closer to OLAP rollups [10].

**Dimension reduction / projections.** Static projections or multi-index layouts differ from *query-time* focus sets over an existing full-dim HNSW. Projected auxiliary indexes (Option 3) remain future work when focus palettes are small and known a priori.

## 7 Discussion

**What to ship.** In-database attribution and hybrid `mask_from_layer` are the positive systems results. Gather kernels are worth keeping as an implementation detail. Verify is useful as an optional correctness mode *with* overfetch on coherent data;  $\alpha$ -blend should not be marketed as improving Option 2.

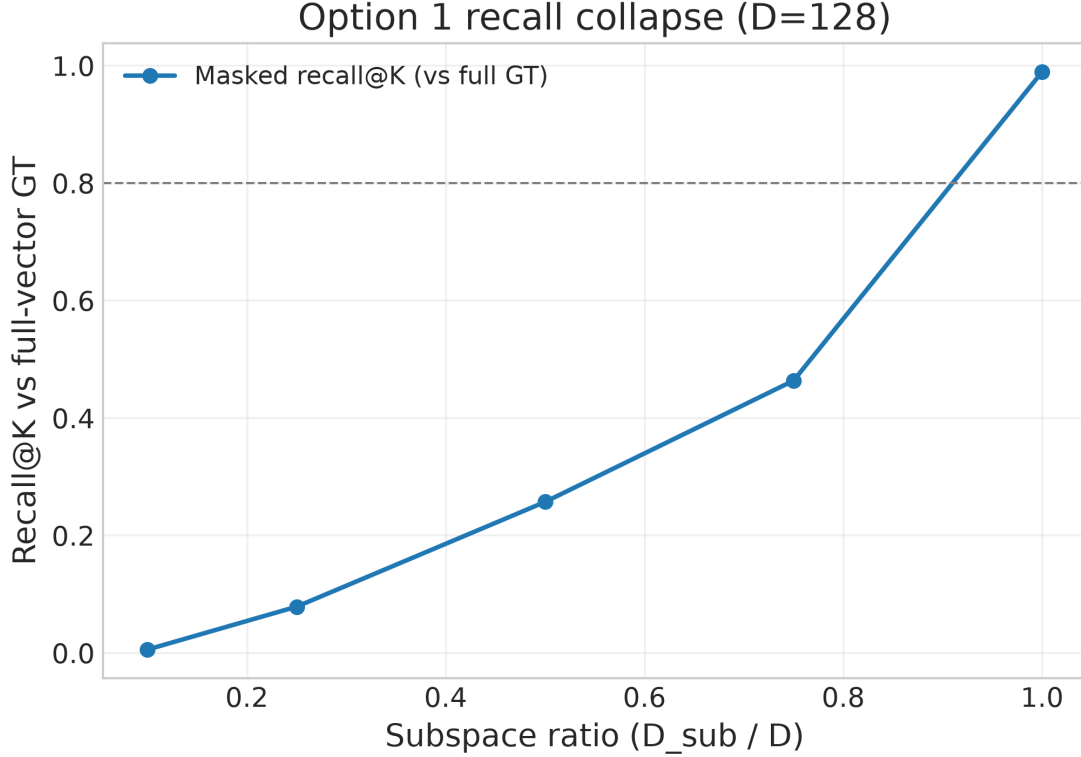


Figure 5: Option 1 on SIFT1M: Recall@10 vs. full-space GT as  $D_{\text{sub}}/D$  shrinks.

Table 1: Keep / demote map for the masked-distance study (canonical runs).

| Knob                 | Verdict     | Evidence                                        |
|----------------------|-------------|-------------------------------------------------|
| Attribution (in-DB)  | Keep        | $\sim 2.2\times$ vs. post-query; recall-neutral |
| Focus rescoring only | Not accel.  | Speedup $< 1$ (Fig. 4)                          |
| Option 1 all-mask    | Baseline    | Full-GT collapse on SIFT (Fig. 5)               |
| Option 2 hybrid      | <b>Keep</b> | Navigability recovery; e2e speedup still $< 1$  |
| X1 gather kernel     | Micro keep  | $1.1\text{--}1.8\times$ ns/op; e2e still no win |
| X2 coherence         | Keep diag.  | Predicts high- $D$ / X3 limits                  |
| X3 verify $m=1$      | API only    | Recall@ $k$ unchanged                           |
| X3 + overfetch       | Conditional | Helps SIFT; $\sim +45$ ms; fails high- $D$      |
| Option 4 $\alpha$    | Skip enh.   | Best at $\alpha=0$ (= Option 2)                 |

**Why no e2e speedup.** Even with faster masked compares, HNSW still touches full vectors for storage, graph structure, and (on hybrid coarse layers) full SIMD. I/O and candidate management dominate the microbenchmarked arithmetic.

**Limitations.** High- $D$  Option 1 lacks a dedicated archive folder (high- $D$  collapse is evidenced via Option 2/X2). We did not instrument visited-set divergence (X4) or build projected indexes (Option 3). Attribution suite hardware differs from the Linux masked-suite host; absolute milliseconds are not cross-compared across suites.

## 8 Conclusion

XQdrant shows that exact feature attribution can live inside a Rust vector engine with negligible quality loss, and that masked-distance HNSW is a viable *navigability* tool when coarse layers keep full distance—but not yet a latency win, and not a substitute for coherent subspaces when

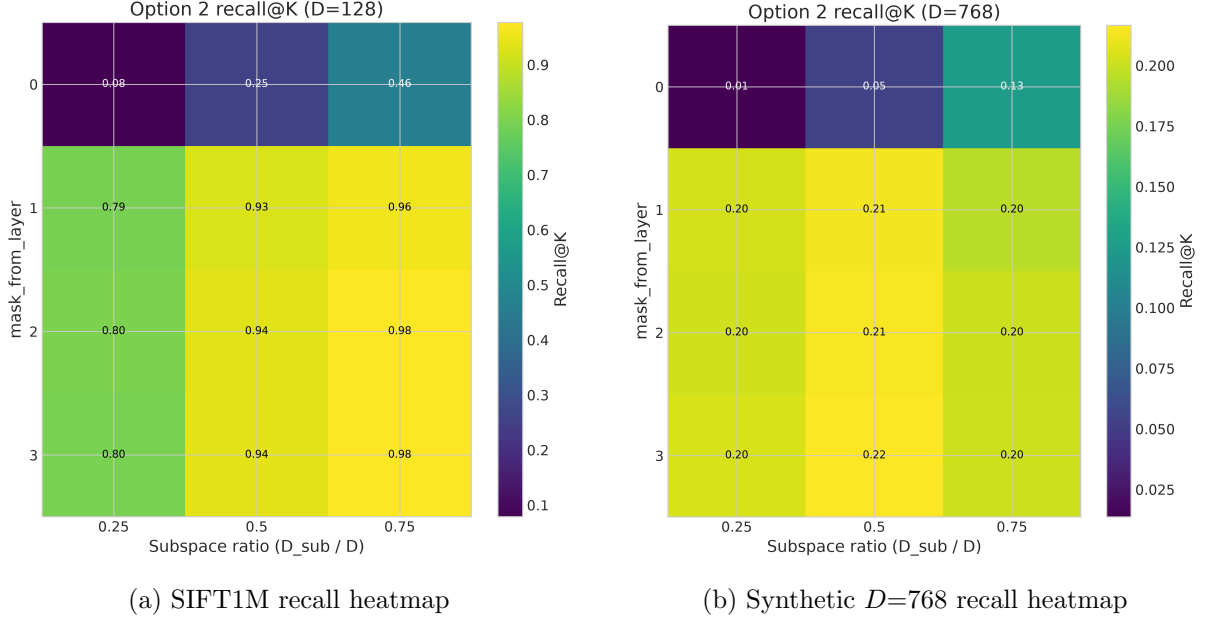


Figure 6: Option 2 (`mask_from_layer`  $\times$  ratio) with gather kernel.

the goal is full-space recall. Overfetch+verify and  $\alpha$ -blending clarify the design space: the former can recover recall when candidates are in range; the latter is dominated by a hard layer cutoff. These measured positives and negatives, backed by a public experiment index, form a concrete basis for subspace-aware vector search in production engines.

## Acknowledgments

We thank the Qdrant open-source community for the engine baseline.

## References

- [1] Yu. A. Malkov and D. A. Yashunin. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. *IEEE TPAMI*, 42(4):824–836, 2020.
- [2] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2021.
- [3] Jianguo Wang et al. Milvus: A purpose-built vector data management system. In *SIGMOD*, 2021.
- [4] Qdrant Team. Qdrant: Vector search engine documentation. <https://qdrant.tech/documentation/>, 2024.
- [5] Pinecone Systems. Pinecone: Managed vector database for production AI. <https://www.pinecone.io/>, 2023.
- [6] Patrick Lewis et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *NeurIPS*, 2020.
- [7] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *NeurIPS*, 2017.

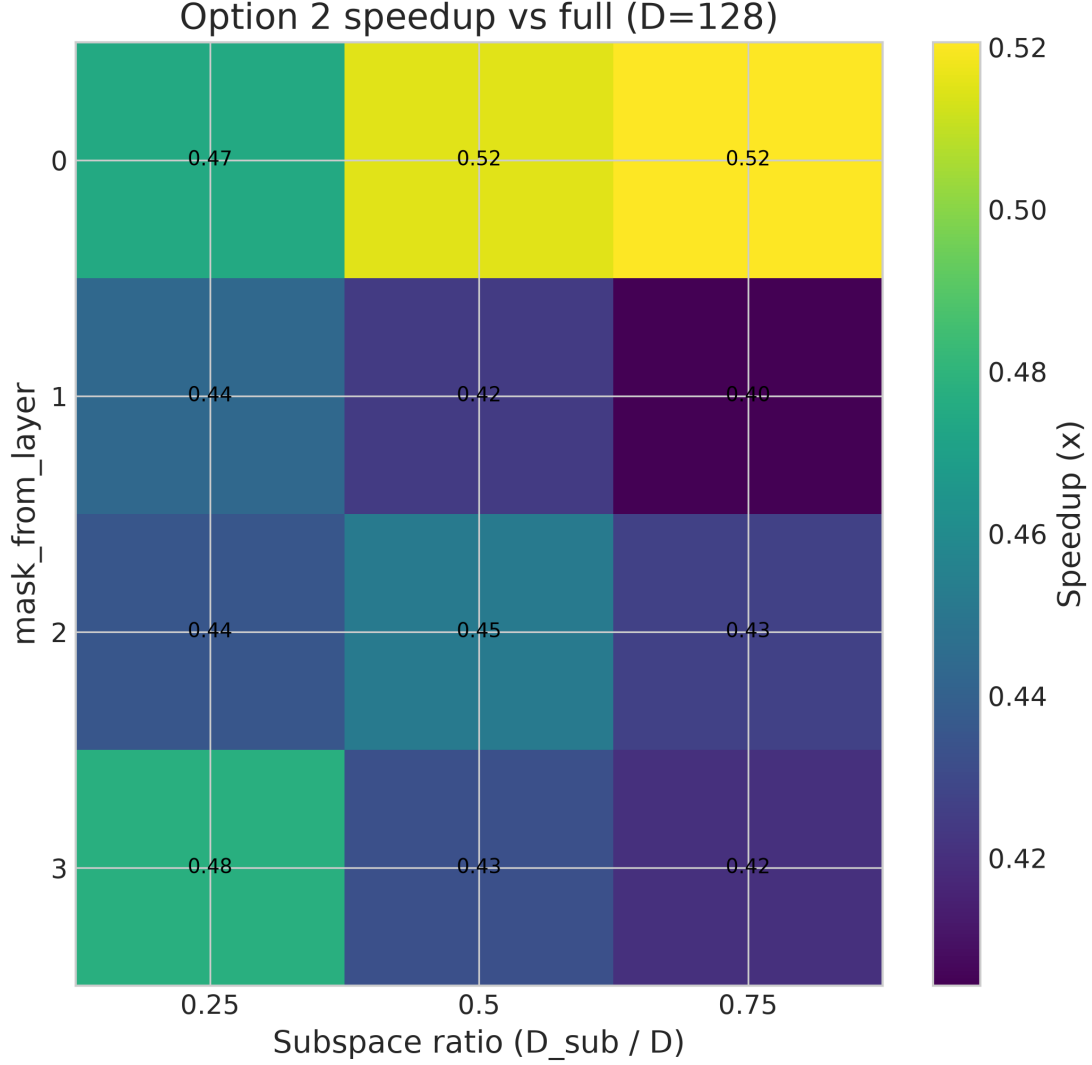


Figure 7: Option 2 SIFT speedup vs. full search: no cell exceeds  $1\times$ .

- [8] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. “Why should I trust you?”: Explaining the predictions of any classifier. In *KDD*, 2016.
- [9] Ruiqi Guo, Philip Sun, Erik Lindgren, et al. Accelerating large-scale inference with anisotropic vector quantization. In *ICML*, 2020.
- [10] Jim Gray et al. Data cube: A relational aggregation operator generalizing group-by, cross-tab, and sub-totals. *Data Mining and Knowledge Discovery*, 1(1):29–53, 1997.
- [11] Hervé Jégou et al. Aggregating local descriptors into a compact image representation. In *CVPR*, 2010. (SIFT descriptor lineage; SIFT1M widely used in ANN evaluation.)
- [12] XQdrant Contributors. XQdrant: Dimension explainability and masked-distance HNSW extensions to Qdrant. Technical report / artifact, 2026. Experiment index: [DB.systems/experiments/README.md](#).

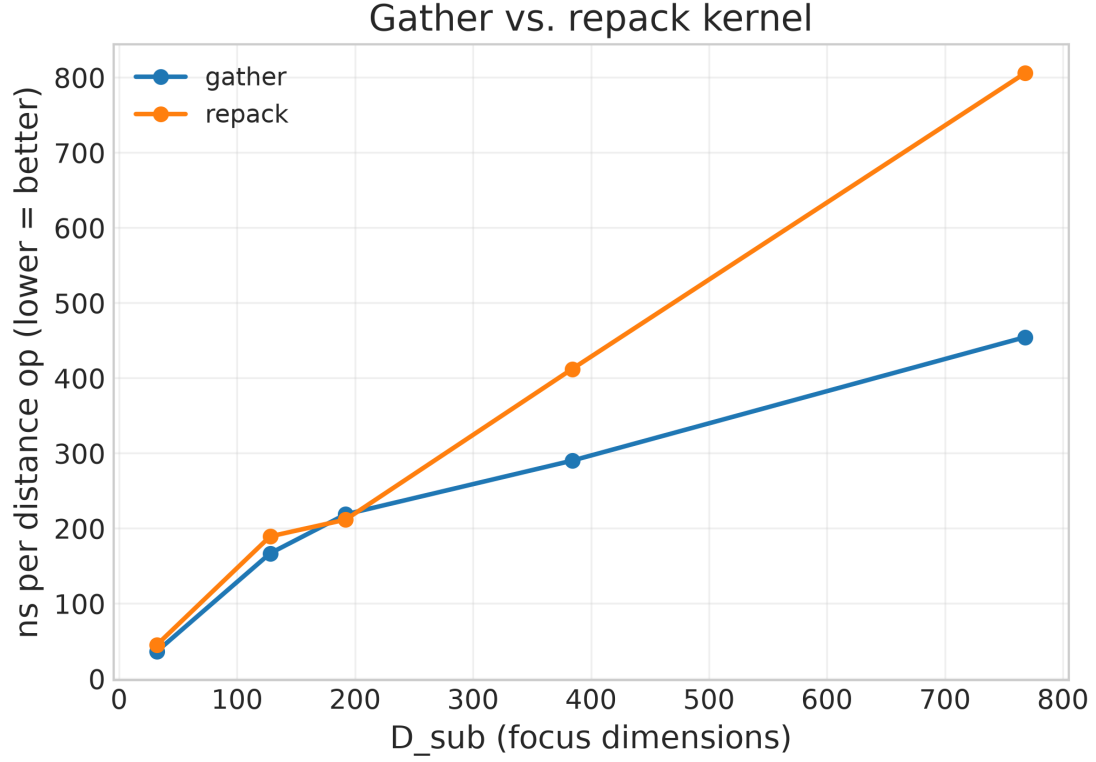


Figure 8: X1: masked Dot kernel throughput (repack vs. gather).

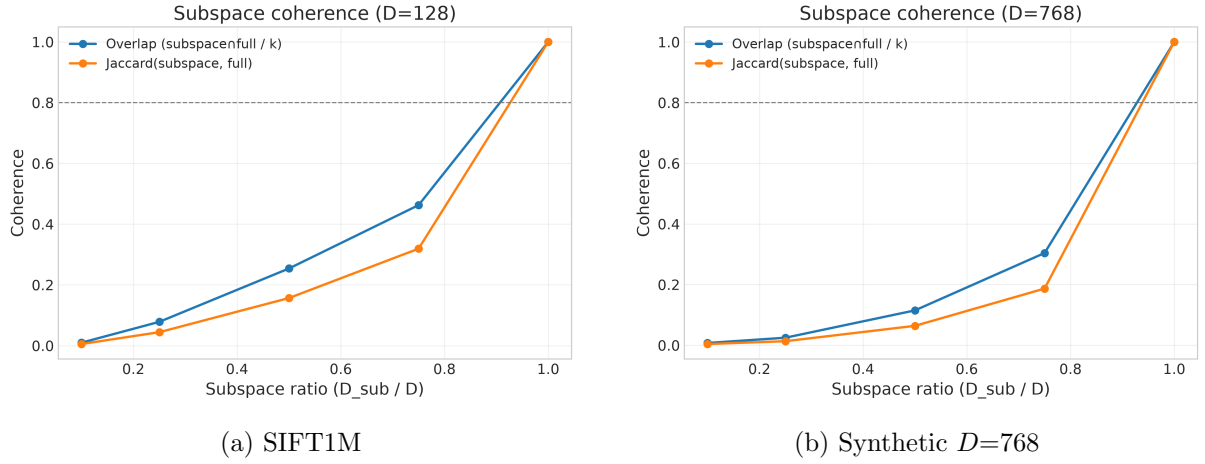
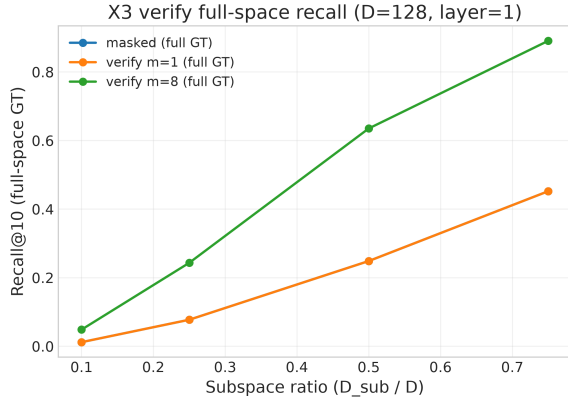
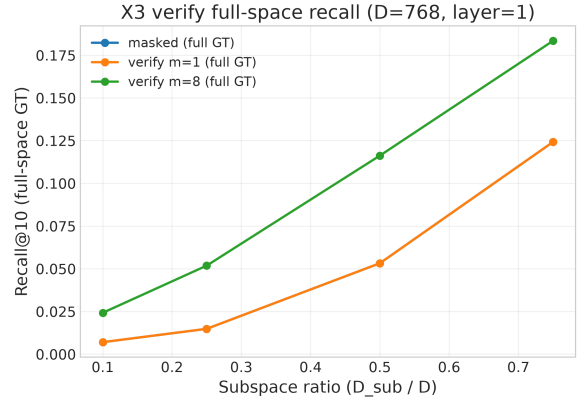


Figure 9: X2: subspace↔full exact top-10 overlap vs. ratio.



(a) SIFT, hybrid  $L=1$



(b) Synthetic  $D=768$ , hybrid  $L=1$

Figure 10: X3: full-space Recall@10 for masked vs. verify at overfetch  $m \in \{1, 8\}$ .

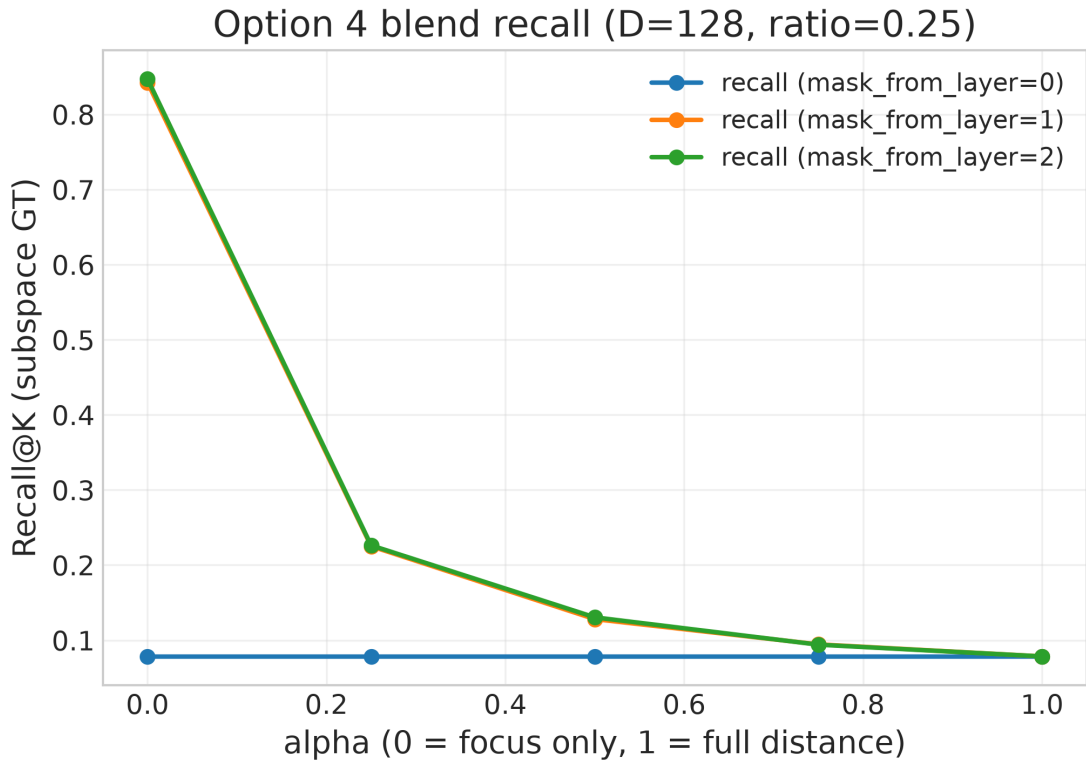


Figure 11: Option 4 on SIFT: recall vs.  $\alpha$  for cutoffs  $\{0, 1, 2\}$  (ratio 0.25, subspace GT).